

Exact Spectrum, Transient Kernel, and Capacity Limits of the M/M/1/K Queue

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Abstract

We give a convention-complete spectral atlas for the finite-capacity continuous-time M/M/1/K birth-death queue. The capacity K includes the customer in service, so the two reflecting endpoint diagonals differ from the interior diagonal. Detailed balance conjugates the generator to a symmetric Jacobi matrix. Its Robin endpoint equations yield every nonzero mode and the closed gap $\gamma_K = \lambda + \mu - 2\sqrt{\lambda\mu} \cos(\pi/(K+1))$. Reassembling the modes gives the exact transient kernel and a reversible total-variation mixing bound. We also treat $K=0$, zero-rate absorbing faces, equal rates, and the three controlled $K \rightarrow \infty$ regimes. The infinite section records only stationary, gap and mass-escape limits; no continuous-spectrum claim is made. Independent exact, symbolic, replay and hostile-mutation audits ship with the theorem.

1 Model and endpoint convention

States are $\{0, 1, \dots, K\}$, with K including the customer in service. For $K \geq 1$ and positive rates,

$$Q_{0,0} = -\lambda, \quad Q_{0,1} = \lambda, \quad Q_{n,n-1} = \mu, \quad Q_{n,n} = -(\lambda + \mu), \quad Q_{n,n+1} = \lambda, \quad (1)$$

for $0 < n < K$, and $Q_{K,K-1} = \mu$, $Q_{K,K} = -\mu$. Thus rows sum to zero. At $K=0$ the generator is the one-by-one zero matrix. Time is the physical continuous-time Markov clock and $P_t = \exp(tQ)$ is row stochastic.

Theorem 1 (Finite M/M/1/K spectral atlas). *Put $\rho = \lambda/\mu$ and $Z_K = \sum_{r=0}^K \rho^r$. For $\lambda, \mu > 0$,*

$$\pi_n = \frac{\rho^n}{Z_K}, \quad 0 \leq n \leq K,$$

and $S = D_\pi^{1/2} Q D_\pi^{-1/2}$ is symmetric tridiagonal with off-diagonal $r = \sqrt{\lambda\mu}$. Its eigenvalues are 0 and

$$\nu_j = -(\lambda + \mu) + 2r \cos \theta_j, \quad \theta_j = \frac{j\pi}{K+1}, \quad j = 1, \dots, K. \quad (2)$$

For $j \geq 1$, an eigenvector is proportional to

$$v_j(n) = \sin((n+1)\theta_j) - \sqrt{\mu/\lambda} \sin(n\theta_j), \quad 0 \leq n \leq K, \quad (3)$$

and $v_0(n) = \sqrt{\pi_n}$. With normalized vectors,

$$P_t(i, j) = \sqrt{\frac{\pi_j}{\pi_i}} \sum_{m=0}^K v_m(i) v_m(j) e^{\nu_m t}, \quad \nu_0 = 0. \quad (4)$$

The spectral gap is

$$\gamma_K = \lambda + \mu - 2\sqrt{\lambda\mu} \cos \frac{\pi}{K+1}, \quad (5)$$

and

$$\|P_t(i, \cdot) - \pi\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\pi_i^{-1} - 1} e^{-\gamma_K t}. \quad (6)$$

Proof. The ratios $\pi_{n+1}/\pi_n = \lambda/\mu$ give detailed balance, hence the displayed similarity and the common off-diagonal r . The interior equation for an eigenvector is the second-order recurrence $rv_{n-1} - (\lambda + \mu)v_n + rv_{n+1} = \nu v_n$. Writing $\nu = -(\lambda + \mu) + 2r \cos \theta$ gives (3) after imposing the left endpoint equation. The right endpoint residual is $(\lambda + \mu - 2r \cos \theta) \sin((K+1)\theta)$, so the nonzero modes have $\theta_j = j\pi/(K+1)$. The constant stationary mode supplies v_0 ; symmetry then gives an orthonormal basis and (4) by conjugating e^{tS} . The smallest positive decay rate is (5). The reversible $L^2(\pi)$ spectral estimate $\chi_i^2(t) \leq (\pi_i^{-1} - 1)e^{-2\gamma_K t}$ and Cauchy–Schwarz yield (6). $\square$

2 What the finite formula buys us

The endpoint convention is mathematically active: replacing $Q_{K,K}$ by the interior diagonal changes the Robin equation and the angle grid. Formula (4) is an entrywise finite sum, so it gives nonnegative, row-stochastic transition probabilities without an unevaluated matrix exponential. It also makes the gap bound auditable for every starting state, including the least likely wall state. At $\lambda = \mu$, π is uniform and (5) becomes $2\lambda(1 - \cos(\pi/(K+1)))$.

3 Singular faces and the capacity boundary

If $K = 0$, state 0 is the sole absorbing state. If $\lambda = 0$ and $\mu > 0$, state 0 absorbs; if $\mu = 0$ and $\lambda > 0$, state K absorbs. When both rates vanish every state is absorbing, so there is no unique stationary law. These faces are not limits of the positive-rate eigenvector formula.

For fixed positive rates, let $K \rightarrow \infty$. If $\rho < 1$, then $\pi_n \rightarrow (1 - \rho)\rho^n$ for each fixed n , while $\gamma_K \rightarrow (\sqrt{\mu} - \sqrt{\lambda})^2$. If $\rho = 1$, the finite law is uniform but no probability stationary law exists on the half-line, and

$$\gamma_K = 2\lambda\{1 - \cos(\pi/(K+1))\} \sim \frac{\lambda\pi^2}{(K+1)^2}. \quad (7)$$

If $\rho > 1$, $\pi_n^{(K)} \rightarrow 0$ for every fixed n : mass escapes toward the reflecting wall, and the half-line chain has no stationary probability. We do not infer a continuous-spectrum decomposition from these elementary limits.

4 Evidence, distinctions, and Route-A boundary

The canonical ledger contains 20 stationary rows, 60 normalized mode rows, 240 kernel rows, 240 linked mixing rows, 16 capacity-limit rows and eight boundary rows. A producer-independent checker makes 3,655 assertions and also performs exact generator/reversibility checks plus a high-precision matrix exponential spot check. SymPy verifies 46 identities, including the Jacobi similarity, endpoint recurrence and Chebyshev characteristic factor. Clean replay is byte-for-byte; 25 repaired-hash semantic/schema mutations and one stale-hash mutation are rejected.

This reversible birth–death semigroup is distinct from C208’s branching probability-generating function and from C220’s interacting-particle matrix-ansatz phase atlas. The strict tuple is

$$(\text{AO_FAIL}, \text{A1_FAIL}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_FORMAL_HINT}),$$

with `ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`. The Jacobi matrix is a finite formal quantization hint only: queue states and rates have no primitive arithmetic owner, target divisor, or Hilbert–Polya bridge.

Source note and declarations

Karlin–McGregor [1] established the classical birth–death spectral framework. Related finite Markov-matrix spectral calculations are discussed by Ekström, Garoni, Jozefiak and Perla [2];

recurrence vocabulary follows Callaert–Keilson [3]. We claim no priority for these results. No target arithmetic, Euler factor, root number, automorphy, target divisor or Route-B input is used. Finite characteristic polynomials are not infinite Fredholm determinants. Generative tools assisted drafting and code generation; the artifact chain checked the displayed claims. Scope literal: NO_BAD_EULER_OR_ROOT_NUMBER. This is not external peer review.

References

- [1] S. Karlin and J. McGregor, “The differential equations of birth-and-death processes, and the Stieltjes moment problem,” *Transactions of the American Mathematical Society* 85 (1957), 489–546. DOI: 10.1090/S0002-9947-1957-0091566-1.
- [2] S.-E. Ekström, C. Garoni, A. Jozefiak and J. Perla, “Eigenvalues and eigenvectors of tau matrices with applications to Markov processes and economics,” *Linear Algebra and its Applications* 627 (2021), 41–71. DOI: 10.1016/j.laa.2021.06.005.
- [3] H. Callaert and J. Keilson, “On exponential ergodicity and spectral structure for birth-death processes, II,” *Stochastic Processes and their Applications* 1(3) (1973), 217–235. DOI: 10.1016/0304-4149(73)90001-X.